Seamless UAV Hyperspectral Image Stitching Using Optimal Seamline Detection via Graph Cuts

Zongyi Peng, Yong Ma, Yujie Zhang, Hao Li, Fan Fan, and Xiaoguang Mei

***Abstract*— Image stitching task targets to derive a large panoramic image for obtaining extensive information. However, artifacts such as ghosting or geometric misalignment are inevitably generated. As a practical measure, optimal seamline detection strategies use the spatial information to obtain the optimal seam in RGB image stitching, but they cannot be directly used in hyperspectral image (HSI) stitching. Since the spatial information of numerous continuous bands of HSI is different, the detected seam of the traditional RGB-based method in each band of HSI is divergent, which will cause visual difference and spectral distortion. To solve this problem, we propose a novel optimal seamline detection strategy via graph cuts for HSI stitching in this work. First, we use robust feature matching and elastic warp to align multiple adjacent images into a common geometrical transformation. After that, we design a novel energy function composing both the spatial and spectral information of HSI to determine an optimal seam in continuous regions with high texture consistency. Finally, we use the graph cuts method to eliminate visible artifacts. Our method can determine a unique optimal seam in the whole HSI for stitching so as to obtain high-quality panoramic HSI without artifacts and reduce the spectral distortion. A series of experiments verify the effectiveness and superiority of the proposed method to several advanced approaches in HSI stitching.**

***Index Terms*— Graph cuts, hyperspectral image (HSI) stitching, optimal seamline detection, spectral similarity.**

I. Introduction

WITH the flourish of hyperspectral remote sensing industries, this project direction has attracted considerable attention in disaster investigation and early warning [1], [2], land-cover mapping [3], and other fields [4], [5]. Compared with RGB and infrared images, hyperspectral images (HSIs) with continuous bands of spectral information have been widespread in many applications [6], [7], [8], [9], [10]. Recently, unmanned aerial vehicle (UAV) system has been thriving into a new hot spot topic for its great lightweight and convenient [11], [12], [13], [14]. However, the field of view of UAVs is limited, and it is hard to gather large-scale environmental information from a single UAV image. Hence, the image stitching technique is meaningful for obtaining panoramic images of a series of adjacent images with overlap regions [15].

Image stitching problem consists of the following scientific technologies: feature matching, image alignment, optimal seamline detection, and image blending. Scale-invariant feature transform (SIFT) [16] and speeded-up robust features (SURFs) [17] are the most typical feature matching methods. After finding the feature points and matching them, the matched features are not all inlier [18], [19]. Consequently, some mismatch removal methods were developed to combat with the outliers, such as random sampling consensus (RANSAC) [20], locality-preserving matching (LPM) [21], vector field consensus (VFC) [22], and locally linear transforming (LLT) [23]. Image alignment technology is a process of spatially evaluating the geometrical transformation of multiple adjacent images based on the feature matching result. AutoStitch [24] uses unified global transformation model to align all the images. To obtain higher stitching accuracy, space-varying transformation methods were developed, the popular methods include adaptive as-natural-as-possible (ANAP) [25] and as-projective-as-possible (APAP) [26]. Although some progress has been made in the research of image alignment, there is no method that can completely overcome the artifacts. Hence, the optimal seamline detection strategy aims to conceal the artifacts by determining a continuous seam that is expected to avoid crossing visible objects in the overlap regions between images [27], [28], [29]. The optimal seamline detection methods decide which pixels should be decided to assign in the subsequent stitched image to minimize visible seam. The optimally detected seamline is a set of pixels with minimal intensity and geometric differences between adjacent images. Image blending is applied to eliminate the large color difference between adjacent images caused by lighting, weather, camera exposure, etc. including average blending (AB) [30], linear weighted blending (LWB) [31], Poisson blending (PB) [32], and Laplacian pyramid blending [33]. If the adjacent images do not have obvious color difference, the image blending step can be omitted.

Nowadays, various methods of optimal seamline detection have been introduced in urban orthophoto maps and RGB images [27], [34], [35], [36], [37], [38]. However, these traditional RGB-image-based methods cannot be applied in HSI stitching directly. Compared with RGB images, HSIs

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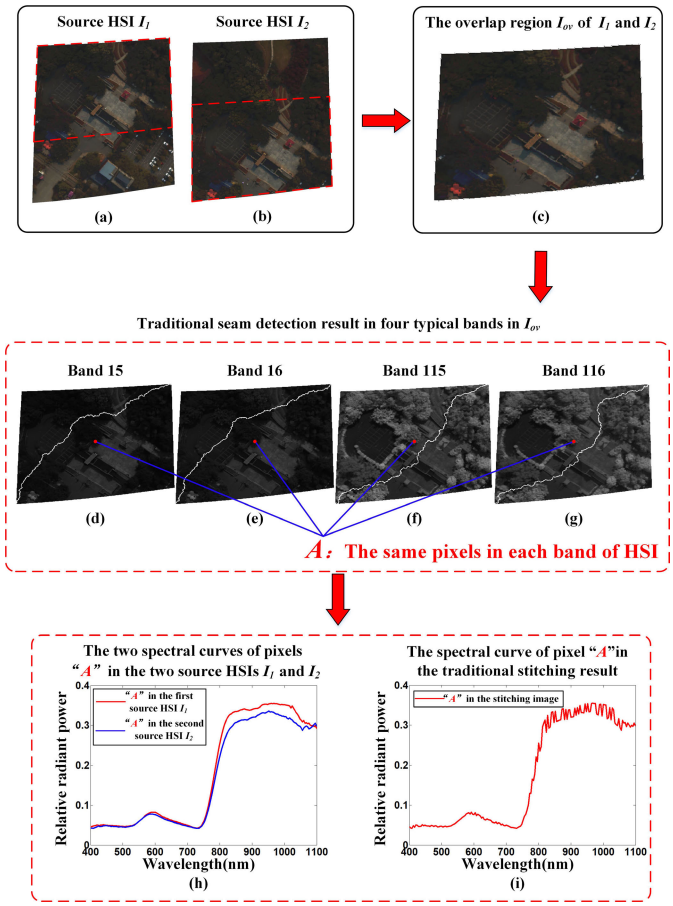


Fig. 1. Description of spectral distortion in the traditional stitching results. (a) and (b) Two aligned input source HSIs. (c) Overlap region in the input two HSIs. (d)–(g) Detected seams of the traditional method in the overlap region in different bands of HSI, and point "A" are the pixels of same position in each band of HSI. (h) Spectral curves of pixels "A" in the first and second source HSIs. (i) Spectral curve of pixel "A" in the traditional stitching result.

consist of multitudinous constant bands, and each pixel can be described as a continuous spectral curve that contains rich spectral information. Since the spatial information in hundreds of bands of HSI is diverse, the detected seam of the traditional methods in each gray-scale band is different [39]. Fig. 1 shows the description of spectral distortion in the traditional stitching result. Fig. 1(a) and (b) shows the two aligned input source HSIs, and the overlap region of two source HSIs is shown in Fig. 1(c). For convenient observation, Fig. 1(a)–(c) shows the pseudocolor images. Fig. 1(d)–(g) gives the detected seams of the traditional RGB-image-based method [40] in the overlap region in different bands of HSI. The spatial information in different bands [i.e., Fig. 1(d)–(g)] of HSI is different, so the detected seams are also different. Even in the adjacent bands [i.e., Fig. 1(d) and (e)], there are still small differences in the detected two seams. As an example, points "A" are the pixels of the same position in each band of HSI. Because the detected seams are different in different bands, one can find that the pixels "A" in Fig. 1(d) and (e) are below the seams, but in Fig. 1(f) and (g) are above the seams. As we know, the optimal seamline detection strategy is implemented by coping each source image to the corresponding side with a determined seam to comprise the stitching result. In this case, it is not hard to find that some pixels near the seam (i.e., such as pixels "A") in different bands will be divided into different source images. The two spectral curves in Fig. 1(h) represent the spectra of pixels "A" in the two source HSIs. Even though the two curves are different because of the HSI acquisition, they are both smooth and can reflect spectral information well. However, as shown in Fig. 1(i), the curve indicates the spectrum of pixel "A" in the traditional stitching result. Because the detected seam in each band of stitching HSI is different, the values of each position of the curve in Fig. 1(i) are derived from the same position of one of the two curves in Fig. 1(h), and we can find that there are many barbs in this curve, suffering serious spectral distortion. In some complex scenes, this phenomenon of different detected seams in HSI also leads to visual differences. Moreover, to the best of our knowledge, there are few researches on HSI stitching up to now, and the spectral information of HSI has not been used for stitching.

To address the aforementioned challenges, we propose a unified graph cuts minimization strategy to obtain a unique perceptually optimal seam in the whole HSI for stitching. To guarantee the high quality of detected seamline, we design a novel energy function that contains both the spatial and spectral information. Not only both the color intensity and gradient magnitude in a single gray-scale band are composed but also the spectral similarity inspired by the self-weighted correlation coefficients (SCCs) of each pixel is considered in the energy function. Our strategy overcomes the problem that the optimal seamline detection method could not be used in HSI stitching directly while magnificently concealing the obvious artifacts caused by image misalignments and ensuring the spectral information consistency of each pixel of HSI. Our work contains several key steps. In the first phase, we apply robust feature matching and elastic alignment to align multiple adjacent images into the same geometrical transformation [41]. This step is the premise of detecting the optimal seamline. Then, we introduce a novel energy function composing both the spatial and spectral information of HSI that combines image color, gradient magnitude, and spectral similarity information to detect an optimal seam. Note that the spatial information is extracted in a single band which has the most peak signal-to-noise ratio (PSNR) estimation, and the spectral similarity information is based on the whole HSI. Finally, the energy function is solved via graph cuts to achieve seamless stitching for HSI.

The primary contributions in this work can be summarized as follows.

1) We introduce a novel optimal seamline detection strategy for HSI stitching, which solves the dilemma that the optimal seamline detection method cannot be used in HSI stitching directly, as well as deriving high-precision panoramic HSI without obvious artifacts while reducing the spectral distortion.
2) We design a new energy function that considers both the spatial and spectral information of HSI, and the graph cuts algorithm is solved to effectively obtain the perceptually optimal seam. To the best of our knowledge, it is the first time that spectral information is considered for the HSI stitching task.

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3) We implement series experiments in several aspects to validate the stitching results on both the spatial and spectral performance, and the qualitative and quantitative results verify the superiority of our method.

II. Related Work

A. Image Stitching

We briefly review the three steps of HSI stitching: feature matching, image alignment, and image blending. Note that the step of optimal seamline detection is the focus of this work, so the review of optimal seamline detection is introduced in Section II-B separately.

Feature matching is the underlying task for image stitching [18], [19], [42]. The popular methods of feature matching are SIFT [16] and SURF [17]. To ensure the accuracy of matching result, the mismatch removal task aims to remove the outlier of the detected feature points. RANSAC [20] is the classical method for removing false matches, but it cannot be applied in some complex situations, such as the motion of the image scene being nonrigid. Hence, some robust nonparametric fitting methods are applicable to more complex nonrigid situations, including LPM [21], VFC [22], LLT [23], mTopKRP [43], etc. Moreover, Mo et al. [44] proposed a novel deep-learning-based feature extraction and matching framework of adjacent HSIs. These feature matching strategies have been verified to be both robust and efficient.

The image alignment step is aimed at evaluating the geometrical transformation of adjacent images based on the feature matching result. Nowadays, many methods have been proposed and achieved well results. The most typical method AutoStitch uses unified global transformation model to align all the images [24], but the result is impressible to parallax. To derive better visual effects, local space-varying transformation methods have been developed. APAP uses local adaptive transformation but still has the distortion issue [26]. Shape-preserving-half-projective (SPHP) overcomes this issue using projective transformations, which is constantly turned to a global parallel between images [45]. Based on the previous studies, adaptive ANAP composes a hybrid transformation, which decreases the visual distortion [25]. Natural image stitching with the global similarity prior (NISwGSP) was proposed by Chen and Chuang [46]. Li and Zhou [47] proposed a quaternion rank-1 alignment model for image stitching. Although the above approaches have achieved good effects in some situations, for irregular cases of unnatural rotation or nonrigid transformation, artifacts such as ghosting or misalignment still emerge.

Image blending is the final step to eliminate the large color difference caused by lighting, weather, camera exposure, etc. AB [30], linear blending [31], [41], PB [32], and Laplacian pyramid blending [33] are known as the most classical methods. When the adjacent images do not suffer obvious color difference, the image blending step can be omitted.

B. Optimal Seamline Detection

The optimal seamline detection methods determine a continuous seam in the overlap regions between a pair of adjacent images where the texture differences are minimal, and these approaches have been widely used for RGB image stitching [28], [29], [35]. For example, the twin snakes [48] algorithm was designed to automatically derive the optimal route of seamline. Dijkstra's algorithm [49] was generally used to solve the case of moving targets. The seam-driven [50] algorithm evaluates the optimal geometrical transformation of adjacent images by the seam-cutting strategy. A fast and robust seam estimation (FARSE) method defines an energy map with gray and gradient information to eliminate the obvious artifacts [40]. Moreover, the graph cuts [35], [51], [52], [53], [54], [55] optimization strategies can availably obtain the ideal image segmentation results with higher efficiency. Hence, the graph cuts method is currently used in many aspects, such as image stitching [27] and graphics segmentation [54]. Although these methods can derive good results in RGB images, they cannot be applied in HSI stitching directly. To the best of our knowledge, there are few researches on HSI stitching, and the optimal seamline detection strategy has not been applied for HSI stitching.

C. Graph Cuts

The graph cuts method has been broadly used in image stitching to solve the problem in terms of energy optimization [51], [52], [53], [54]. Specifically, the energy optimization step is described as a binary labeling definition task. p is an arbitrary element in the elements set $\mathbf{P}$, $\mathcal{N}$ is the adjacent element pairs set in the neighborhood, and $\mathbf{L}$ is a label set. The task of this approach is to derive a labeling l (i.e., a map from $\mathbf{P}$ to $\mathbf{L}$) by minimizing the energy cost function

$$E(l) = \sum_{p \in \mathbf{P}} D_p(l_p) + \sum_{(p,q) \in \mathcal{N}} S_{p,q}(l_p, l_q) \tag{1}$$

where (p, q) is a pair of neighboring elements over $\mathcal{N}$. The data term $D_p(l_p)$ reflects the cost of attaching the label l_q to the element $p \in \mathbf{P}$, and the smoothness term $S_{p,q}(l_p, l_q)$ denotes the cost of selecting pair labels (l_p, l_q) to the element $(p, q) \in \mathcal{N}$. Ideally, if the labels l_p and l_q have the same value, the smoothness term $S_{p,q}(l_p, l_q)$ is expected to be equal to 0.

Nowadays, the graph cuts strategy has been widely used to determine the optimal seam. However, the traditional energy function only considers the color difference, which is not suitable for many complex scenes. Moreover, more image texture information such as gradient magnitude and texture complexity are considered to determine the seam [27], [35]. Although these methods have achieved good results in RGB images, they cannot be directly used for HSI stitching. Because the spatial information of multitudinous bands of HSI is different, the detected seam of the traditional optimal seamline detection methods in every single band of HSI is divergent, leading to visual difference and spectral distortion [41]. Hence, the current problem is how to consider the distinctive spectral information of the HSI in the energy function to derive a perceptually optimal seamline directly in the whole HSI.

III. Proposed Method

The several steps of our proposed method for HSI stitching are introduced in detail. Our goal is to obtain the stitched

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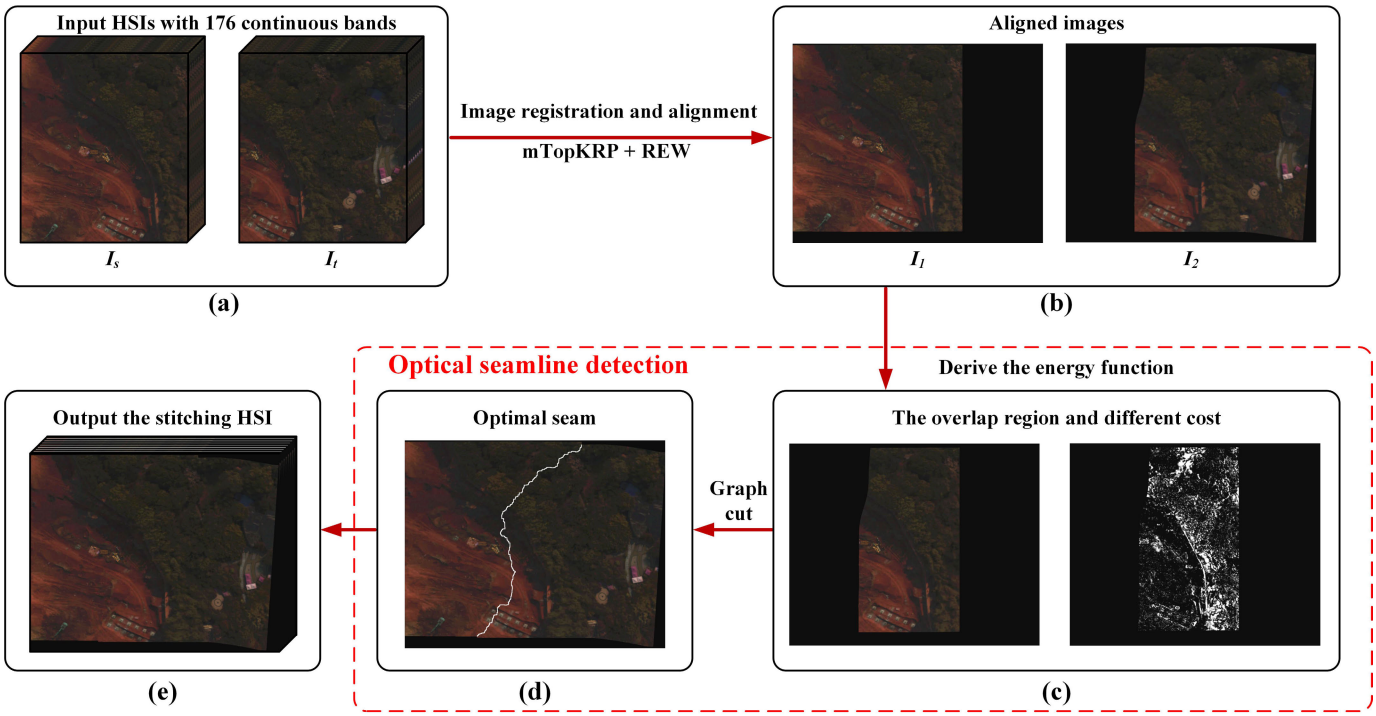


Fig. 2. Flowchart of the proposed method. (a) Input HSIs. (b) Aligned images. (c) Overlap region and the difference cost. (d) Optimal seam. (e) Stitching result.

hyperspectral panoramic image without artifacts such as visible seams and ghosting while ensuring the consistency of the spectral information of stitching HSI. We first briefly describe the feature matching and image alignment processes. These steps align multiple adjacent images into a common coordinate system, which are the premises of determining the optimal seamline. Second, we define an innovative energy function, which fuses the information of color intensity, gradient magnitude, and distinctive spectral similarity of HSI. Finally, we apply the graph cuts approach to derive the optimal seam. The flowchart of our method is illustrated in Fig. 2. It should be mentioned that the image blending step is used to eliminate the large color difference between adjacent images caused by lighting, weather, camera exposure, etc. For the HSI collected by UAV, most of the HSI data are collected at one time, and the collection process is shorter than that of satellite HSI. In this case, there is no obvious color difference between adjacent HSIs. Moreover, to protect the spectral information of the source HSIs, the image blending step has been omitted in our proposed method.

A. Feature Matching and Image Alignment

In this article, for the adjacent HSIs with overlap regions, we first derive the matching result based on the SIFT [16] and mTopKRP algorithms [43] and then construct robust elastic warp [41]. These processes ensure local alignment in the overlap region even if the images suffer from scale transformation, irregular rotation, or other complex cases. Our feature matching and image alignment strategies are not only suitable for the simple case of two adjacent images but also effective in overcoming the cumulative errors of multiple images. By performing this strategy, we obtain multiple HSIs in the same coordinate system.

B. Optimal Seamline Detection

After the image alignment, the adjacent images have been warped in the same coordinate system as precisely as possible. However, due to the existence of parallax, artifacts such as geometric misalignments and ghosting inevitably occur in the overlap region. Therefore, we use the optimal seamline detection strategy to copy each aligned original image to the corresponding side to eliminate artifacts. The traditional energy definition task for RGB image stitching tends to use Euclidean-metric color difference. Moreover, more image information has been considered, such as gradient magnitude and texture complexity. These methods have been developed in RGB image stitching and achieved well results, but they are unsuitable for HSI stitching. To settle this problem, we define a novel energy cost function by taking the spatial and spectral information of HSI into account simultaneously, which can determine a unique optimal seamline in HSI directly.

1) Energy Definition: For a component image $\mathcal{I}$ from two aligned original images I_1 and I_2, the energy cost $C(\mathbf{x})$ of each pixel $\mathbf{x} = (x, y)$ in $\mathcal{I}$ consists of three terms: the color difference term $C_c(\mathbf{x})$, the gradient magnitude term $C_g(\mathbf{x})$, and the spectral similarity information $C_s(\mathbf{x})$, i.e.,

$$C(\mathbf{x}) = C_c(\mathbf{x}) + C_g(\mathbf{x}) + \varepsilon C_s(\mathbf{x}). \tag{2}$$

In the above equation, the items $C_c(\mathbf{x})$ and $C_g(\mathbf{x})$ represent the spatial information, the item $C_s(\mathbf{x})$ represents the spectral information of HSI, and ε is the tradeoff factor to balance the influence of the spatial and spectral information of HSI. Since the spatial information of hundreds of bands of HSI is different, a specific band with the most PSNR index among the hundreds of bands of HSI is selected to exploit the spatial information [i.e., the first two items in (2)]. The color

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difference is the most common information to be considered to determine the optimal seamlines for image stitching. The color difference term $C_c(\mathbf{x})$ is obtained by

$$C_c(\mathbf{x}) = |I_1(\mathbf{x}) - I_2(\mathbf{x})| \tag{3}$$

where $I_1(\mathbf{x})$ and $I_2(\mathbf{x})$ represent the intensity values of pixel $\mathbf{x}$ in the selected single band with the most PSNR index in the HSI.

Moreover, the energy term $C_g(\mathbf{x})$ is described to distinguish the obvious large gradient differences' objects such as buildings and cars, and the definition of energy term $C_g(\mathbf{x})$ is determined as follows:

$$C_g(\mathbf{x}) = \frac{1}{4}\left(\left|G_1^x(\mathbf{x})\right| + \left|G_2^x(\mathbf{x})\right| + \left|G_1^y(\mathbf{x})\right| + \left|G_2^y(\mathbf{x})\right|\right) + \left|G_1^x(\mathbf{x}) - G_2^x(\mathbf{x})\right| + \left|G_1^y(\mathbf{x}) - G_2^y(\mathbf{x})\right| \tag{4}$$

where $G_1^x(\mathbf{x})$ and $G_1^y(\mathbf{x})$ describe the horizontal and vertical gradient magnitudes of $\mathbf{x}$ in I_1, respectively, which are defined based on the Sobel operator, and the same definition is used for $G_2^x(\mathbf{x})$ and $G_2^y(\mathbf{x})$.

The above-defined two energy terms considered spatial information in the highest PSNR index band of HSI. However, the spatial information of multitudinous bands of HSI is diverse; if we only consider the spatial information, the derived seam in each single band is different, which will lead to visual difference and spectral distortion. Therefore, based on the distinctive spectral property of the HSI, we propose a novel energy function that includes both the spatial and spectral information of the HSI. The spatial information is considered by the most PSNR index band among the hundreds of bands of HSI, and the spectral similarity information is determined by the whole HSI.

In our strategy, the spectral similarity information of pixel $\mathbf{x}$ is defined based on the SCC [56], which has been widely and successfully used in spectral matching and land-cover detection [57], [58], [59], [60]. The definition of SCC is expressed as

$$\mathrm{SCC} = \frac{\sum_{i=1}^{n} w_i \cdot (s_i - \bar{s}) \cdot (r_i - \bar{r})}{\sqrt{\left[\sum_{i=1}^{n} w_i \cdot (s_i - \bar{s})^2\right]\left[\sum_{i=1}^{n} w_i \cdot (r_i - \bar{r})^2\right]}} \tag{5}$$

where n is the number of bands, s_i and r_i are the sample spectrum and reference spectrum of the pixel $\mathbf{x}$, respectively, w_i is the weight array, and $\bar{s} = (\sum w_i \cdot s_i)/\sum w_i$ and $\bar{r} = (\sum w_i \cdot r_i)/\sum w_i$ are weighted means of the sample spectrum and the reference spectrum, respectively. To obtain an appropriate weight array, the weight array $\mathbf{w}$ is considered by simulating visual inspection and is defined as

$$\mathbf{w} = \frac{|\mathbf{r}|}{\mathbf{1} + \mathbf{d}}. \tag{6}$$

From (6), we can find that only the values of positions in the reference spectrum are relatively large, and the corresponding weight values will increase. This is consistent with the method of distinguishing and comparing the features in the two spectra in previous studies [61], [62]. Moreover, in the positions of reference spectrum with relatively small values, or the differences between two spectra are relatively large, the corresponding weight values are low. This phenomenon is also predictable that indicate the attention to these points is reducing because of the relatively weak feature. The array $\mathbf{d}$ means the differences in each pixel $\mathbf{x}$ between two spectra and is described by

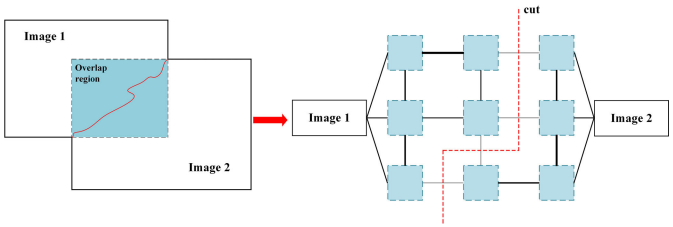


Fig. 3. Illustration of optimal seam via graph cuts. The thickness of lines between adjacent pixels denotes the value of the energy cost. The "cut" marked in red dotted line represents the min-cut, which means the detected optimal seam.

$$\mathbf{d} = |\mathbf{s} - k\mathbf{r}| \tag{7}$$

where k is a factor that diminishes the influence of large difference which is defined by least-squares manner as

$$k = \left(\mathbf{r}^t\mathbf{s}\right)/\left(\mathbf{r}^t\mathbf{r}\right). \tag{8}$$

Depending on the above definition of spectral similarity information, the spectral similarity term $C_s(\mathbf{x})$ for the pixel $\mathbf{x}$ is described as

$$C_s(\mathbf{x}) = \arccos(\mathrm{SCC}). \tag{9}$$

The "arccos" means the inverse cosine of the inverse trigonometric function. This strategy can give ideal performance even if the spectra are noisy or degraded with some interferences [56]. In this way, we can adopt $C_s(\mathbf{x})$ to distinguish the regions with the poor spectral similarity between pixels of adjacent images while not influencing the other context regions with rich edges as used in (2).

2) Energy Function Solution: In our method, the seam detection task is regarded as an energy optimization problem, which can be figured out by the graph cuts approach. As shown in Fig. 3, the derived min-cut indicates the detected optimal seam in the two-label case. In a case of pair warped adjacent images I_1 and I_2, p is an arbitrary pixel in the stitched image. The overlap region of the pair images (I_1 and I_2) is marked with $\mathbf{P}$ and $\mathbf{L} = \{1, 2\}$ is the label set, then the derived seam represents attaching a label $l_p \in \mathbf{L}$ to each pixel $p \in \mathbf{P}$, where the numbers "1" and "2" correspond to I_1 and I_2, respectively. The objective of this approach is to obtain the labeling l (i.e., the map from $\mathbf{P}$ to $\mathbf{L}$) by minimizing the energy function

$$\widetilde{E}(l) = \sum_{p \in \mathbf{P}} \widetilde{D}_p(l_p) + \delta \sum_{(p,q) \in \mathcal{N}} \widetilde{S}_{p,q}(l_p, l_q) \tag{10}$$

where δ is a weight coefficient, $\mathcal{N}$ is a four-connected neighborhood system of pixels, and (p, q) is a pair of neighboring pixels in $\mathcal{N}$. The data term $\widetilde{D}_p(l_p)$ indicates the cost of attaching the label l_q to the element $p \in \mathbf{P}$, it and is defined as

$$\widetilde{D}_p(l_p) = \begin{cases} 0, & p \in \text{overlap} \\ \lambda, & \text{otherwise} \end{cases} \tag{11}$$

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Algorithm 1 HSI Stitching Using Optimal Seamline Detection via Graph Cuts

Input: Two aligned adjacent hyperspectral images I_1 and I_2 with L-bands

Output: Panoramic hyperspectral image without artifacts

1: Select a specific band with the most peak PSNR estimation among the hundreds of bands of HSI
2: In the specific band, derive the overlap region O of two aligned images
3: Obtain the spatial information in the overlap region O using (3) and (4)
4: **for** $j = 1$ to L **do**
5: Obtain the difference array $\mathbf{d}$ between two spectra of each pixel $\mathbf{x}$ in the overlap region O using (7) and (8)
6: Calculate the appropriate weight array $\mathbf{w}$ by (6)
7: Derive the self-weighted correlation coefficient of each pixel $\mathbf{x}$ in the overlap region O based on (5)
8: Obtain the spectral similarity for the pixel $\mathbf{x}$ by (9)
9: $j = j + 1$
10: **end for**
11: Calculate the energy function of each pixel $\mathbf{x}$ by (2)
12: Minimize the energy function by graph cuts using (10)–(12)
13: Derive the perceptually optimal seam in the overlap region O of two aligned HSIs and finish stitching

where λ denotes a very large penalty to ensure the seam is located in the overlap region $\mathbf{P}$. This term means that the pixels in the overlap region are treated equally. Conversely, the pixels in the nonoverlap region are penalized.

The smoothness term $\widetilde{S}_{p,q}(l_p, l_q)$ describes the cost of attaching a pair of labels (l_p, l_q) to the item $(p, q) \in \mathcal{N}$; if the labels of neighboring pixels p, q are diverse [i.e., $(l_p) \neq (l_q)$], it means the derived seam crosses the pixels p and q. According to the above-introduced definition, our proposed smoothness term for the two neighboring pixels $\mathbf{x}$ and $\mathbf{y}$ with the sum of their respective costs can be represented as follows:

$$\widetilde{S}_{p,q}\left(l_p, l_q\right) = |l_p - l_q| \cdot (C(\mathbf{x}) + C(\mathbf{y})) \tag{12}$$

where $C(\mathbf{x})$ and $C(\mathbf{y})$ are the energy costs of a pair of neighboring pixels $\mathbf{x}$ and $\mathbf{y}$ expressed in (2). Thus, for the obtained energy function, we can solve it through the graph cuts model and derive the perceptually optimal seam between two warped adjacent HSIs. To better illustrate the stitching process, we show the algorithm flowchart in Algorithm 1. It should be mentioned that this algorithm flowchart focuses on the proposed optimal seamline detection step of HSI stitching, and the definitions of feature matching and image alignment steps can be found in our previous work [41].

C. Complexity Analysis

So far, we have constructed the energy function with both the spatial and spectral information and used the graph cuts to derive the optimization solution. In the process of constructing the energy function, the first step is to calculate the spatial information with the most PSNR index band among the L-bands of HSI. This step is simple and the computational complexity can be omitted. After that, the spectral similarity information is derived by analyzing the difference between two spectral curves of each pixel in the overlap region in adjacent HSIs. The computational complexity of deriving the spectral information is $O(Ln)$, where n is the number of pixels in the overlap region. For the input HSIs of size $X \times Y$ with the L-bands, the designed energy function is presented as a map and minimized by the graph cuts model, and the computational complexity of optimization will be $O(XY)$. In a word, the overall computational complexity of our method is $O(Ln+XY)$. Compared with the traditional HSI stitching method that only considers the spatial information, although our proposed method using both the spatial and spectral information needs more computational complexity, our approach can achieve better stitching performance and also provides a novel inspiration for the utilization of spectral information of HSI for stitching.

IV. Experiments and Discussions

In this section, the stitching performance of our proposed method is verified by several aspects experiments on real UAV HSIs. The HSI dataset consists of 18 HSIs, the size of which are 960 × 1057. This HSI dataset is captured by a GaiaSky-mini push-broom airborne sensor produced by SiChuan Dualix Spectral Imaging Technology Company Ltd., and the HSIs' data have 176 bands with the spectral range covering 400–1000 nm. This dataset contains various land covers, including trees, bushes, roads, buildings, etc. Moreover, there are a series of typical issues during the collection of the data, such as scale transformation, heavy noise, irregular rotation, and nonrigid transformation. Therefore, experiments on this dataset are able to verify the stitching performance and effectiveness in the real-world cases and applications. All the HSIs are shown in Fig. 4.

At first, we introduce some implementation details. In this article, we first derive the matching result based on the SIFT and mTopKRP algorithms and then construct robust elastic warp to align images. The detailed interpretation of this step can be found in [41]. Moreover, we set the tradeoff factor ε in (2) as 1.5 to balance the influence of the spatial and spectral information of HSI. The value of weight coefficient δ in (10) is assigned as 2 to segment the image in the overlap region more accurately by the graph cuts. All the parameters are set as optimal values to obtain the ideal stitching result.

We verify the effectiveness of our method in a series of experiments. First, the spatial performance of our method is compared with several state-of-the-art methods. Then we contrast with other advanced optimal seamline detection approaches. Next, we analyze the effect of the smoothness term with different energy functions. Finally, we verify the spectral information consistency of ours and several popular image blending methods.

A. Spatial Performance Validation

Five state-of-the-art stitching approaches are applied for comparison with our proposed method, including ANAP [25],

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Fig. 4. HSI dataset for experiments was gathered by a GaiaSky-mini sensor produced by SiChuan Dualix Spectral Imaging Technology Company Ltd., which is a UAV-borne hyperspectral sensor whose wavelength ranges from 400 to 1000 nm. The exhibited pseudocolor images are synthesized by selecting the 92th (700.2 nm), 47th (547.6 nm), and 13th (436.5 nm) bands.

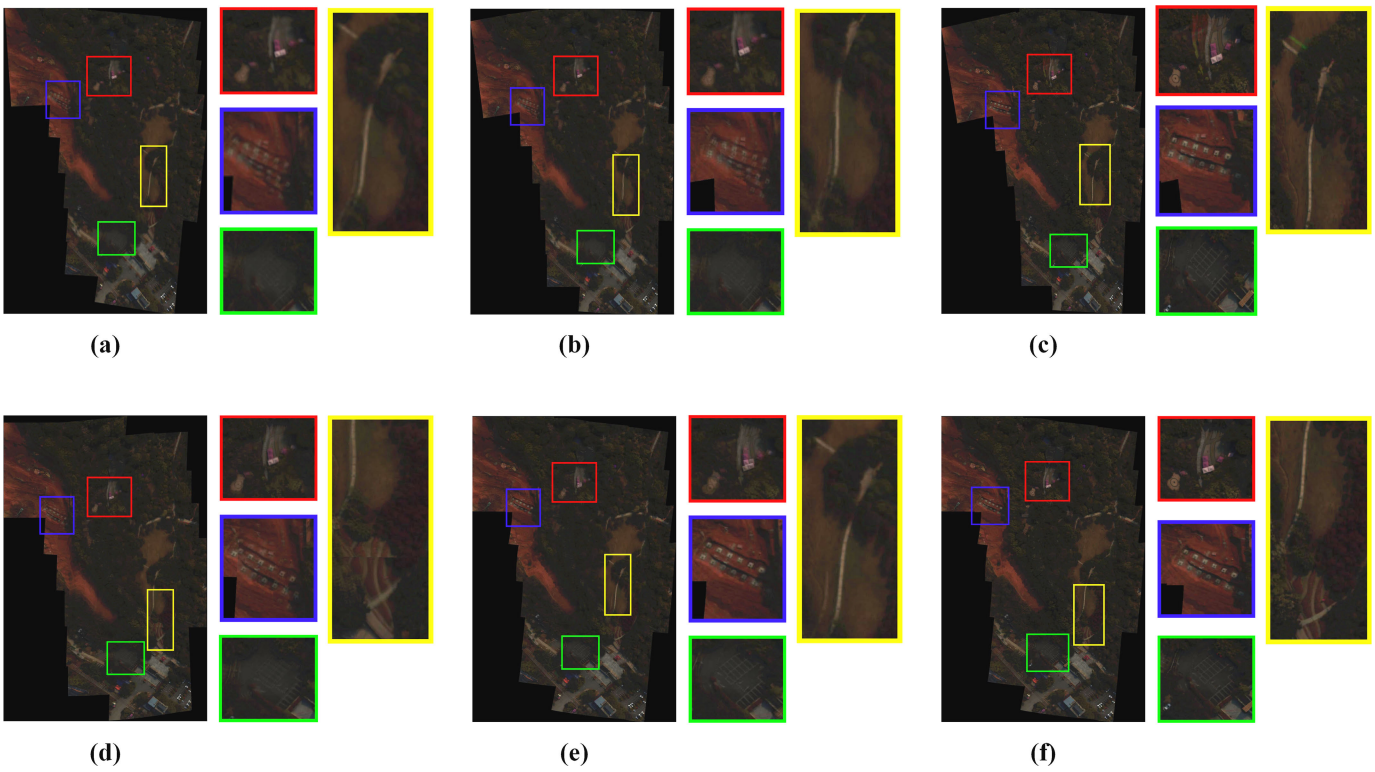


Fig. 5. Qualitative comparison of the spatial stitching performance of our method and state-of-the-art stitching methods by pseudocolor stitched image. (a) ANAP [25]. (b) SPHP [45]. (c) DHW [31]. (d) NISwGSP [46]. (e) REW [63]. (f) Ours.

SPHP [45], DHW [31], NISwGSP [46], and REW [63]. ANAP estimates a combination transformation composed by local homography and global transformation. SPHP uses projective transformations, which is constantly turned to a global parallel between images. DHW adopts dual-homography transformation to warp panoramic images with the remote planes. NISwGSP uses the local alignment and warps each local image by a grid mesh. REW proposes a local transformation approach by a specific function to remove the visual distortion. Note that all the parameters of the compared methods are tuned depending on the default values in the public codes provided by the authors. By incorporating our multiple image stitching method directly into our evaluation, we express the superiority of our strategy and show more details. First, all the HSIs in our dataset are aligned into the same coordinate system by our alignment strategy. Then we detect the optimal seamline in each pair of adjacent warped HSIs to eliminate the artifacts.

The various spatial stitching results are shown in Fig. 5. For convenient and clear observation, these stitching results are comprised as pseudocolor images by extracting 92th (700.2 nm), 47th (547.6 nm), and 13th (436.5 nm) bands in our dataset; the same is true for the rest of the pseudocolor images in this work. One can see from Fig. 5 that our stitching result expresses the best visual effect with the most precise image texture features. Moreover, several positions of typical ground texture are displayed with different color rectangles to derive more details of the local region. In ANAP [25] and SPHP [45], all the enlarged rectangular areas sustain artifacts, especially in the blue rectangle, one can find that there exist ghosting and blurring in the white tiles significantly. In NISwGSP [46], except that the above issues still occur, obvious striping effects appear at the frontier of the overlap region due to geometric misalignment in blue and yellow rectangular local areas. The visual effects in DHW [31] and REW [63] are more precise than in ANAP [25] and SPHP [45], but the blurring or

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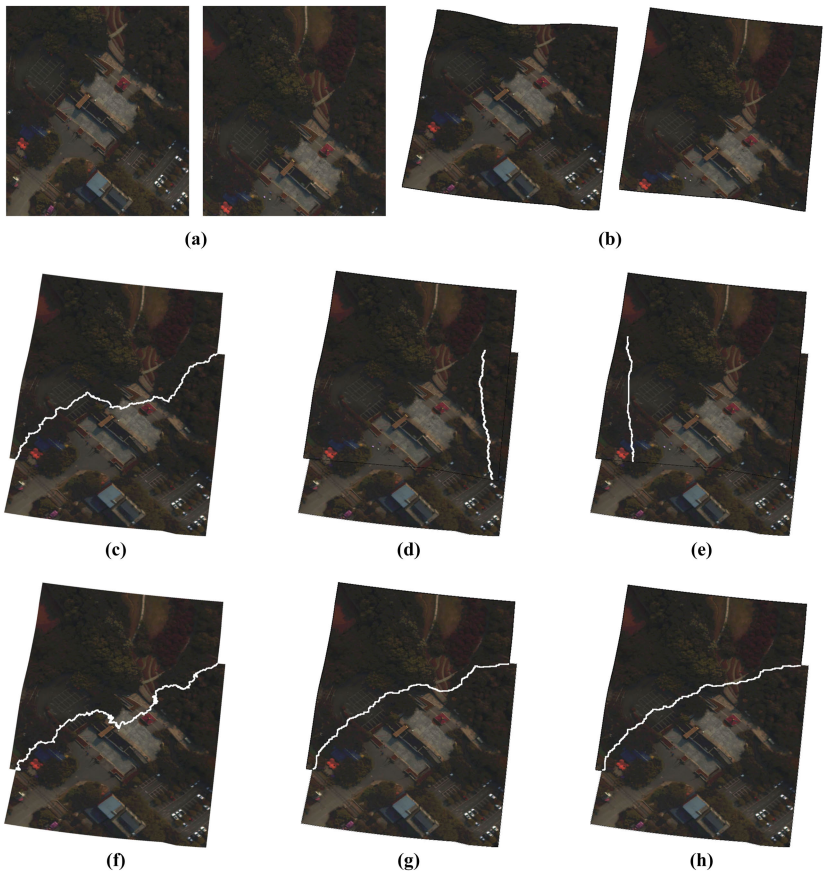


Fig. 6. Seam quality compared with other seamline detection methods. All the detected seams are estimated by the SSIM index quantitatively. (a) Input images. (b) Warped images. (c) Seam-driven [50], SSIM index is 0.9098. (d) DP [64], SSIM index is 0.8887. (e) EDP [65], SSIM index is 0.8717. (f) Farse [40], SSIM index is 0.9195. (g) 5/2 normed [55], SSIM index is 0.9203. (h) Ours, SSIM index is **0.9371**.

repetitions in red and yellow rectangles remain. From the stitching result with the same location in Fig. 5(f), one can find that the image is smoother and clearer. Moreover, as shown in the enlarged local areas in our stitching result, we can observe that our method overcomes all the above issues and achieves a good result. The stitching results illustrate that our method is capable of deriving more natural and accurate visual performance without artifacts in the overlap region.

B. Seam Comparison

We make the seam comparison with several advanced optimal seamline detection methods to verify the superiority of our seam searching method in HSIs. Considering the fairness of the experiments, all the methods depend on the same warped images. As shown in Fig. 6(b), a pair of adjacent images (i.e., the eighth and ninth images) with abundant terrain context are selected in our dataset for the seam comparison experiments. Specifically, we make comparisons with the seam-driven method [50], dynamic programming (DP) method [64], an effective modified seam selection strategy [65] based on DP (using EDP to indicate here), FARSE method [40], and 5/2 power-normed method [55]. The seam-driven method uses the seam-cut strategy to obtain the appropriate geometric transform. The DP method searches the optimal seam in the rectangular overlap regions with efficiency, and the EDP method is the effective modified seam selection strategy based on DP. The FARSE method defines an energy map with gray and gradient information to eliminate the obvious artifacts. The 5/2 power-normed method adopts a better 5/2 power of the L_1-norm to distinguish the weight of color information of images to make the seam avoid significant artifacts. For convenient and clear observation, these results are expressed as pseudocolor images the same as above.

Although some results can be distinguished by human observation, after obtaining the seam result by different seam selection methods, the estimation of the derived seam quality is essential. Counting the number of visible objects crossed by the derived seam is the common strategy in urban orthophoto maps or some RGB image datasets. However, this strategy is not suitable for general situations and is subjective. Hence, to estimate the quality of the detected seam more reasonably and objectively, we adopt the structural similarity (SSIM) index [66] to quantitatively estimate the qualities of derived seams of our proposed method and several advanced approaches. Let $\mathbf{S}$ be the selected seam between a pair of images $\mathbf{I}_1$ and $\mathbf{I}_2$, and $\mathcal{I}$ is the composite image from $\mathbf{I}_1$ and $\mathbf{I}_2$ depending on the detected seam $\mathbf{S}$. The quality measurement SSIM ($\mathcal{I}$) of the seam is expressed as

$$\mathrm{SSIM}(\mathcal{I}) = \frac{1}{N}\sum_{j=1}^{N}\left(\mathrm{SSIM}_i\left(A_j, B_j\right)\right), \quad i = 1, 2 \tag{13}$$

where $\mathrm{SSIM}_i(A_j, B_j)$ denotes the SSIM index of two local regions A_j and B_j from two images I_i and $\mathcal{I}$, respectively, A_j

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and B_j are the image blocks at the jth local region centered at the jth point of $\mathbf{S}$, and N is the total point number of the seam $\mathbf{S}$. The SSIM index itself is determined as a combination of luminance, contrast, and structure components as

$$\mathrm{SSIM}(A, B) = l(A, B)^{\alpha} \cdot c(A, B)^{\beta} \cdot s(A, B)^{\gamma} \tag{14}$$

where $l(A, B) = (2\mu_a\mu_b + C_1)/(\mu_a^2 + \mu_b^2 + C_1)$, $c(A, B) = (2\sigma_a\sigma_b + C_2)/(\sigma_a^2 + \sigma_b^2 + C_2)$, $s(A, B) = (\sigma_{ab} + C_3)/(\sigma_a\sigma_b + C_3)$, μ_a, and μ_b are the mean luminance values in the local regions A and B, respectively, and σ_a and σ_b are the standard variances in A and B, respectively. σ_{ab} is the covariance between regions A and B. Here, C_1–C_3 are small constants to prevent the equation denominator being zero, and α, β, and γ are the balancing coefficients to regulate the proportion of three items. Based on the definition in [66], the following acquiescent settings are used: $C_1 = (0.01 \times L)^2$, $C_2 = (0.03 \times L)^2$, $C_3 = C_2/2$, $L = 255$ for images of the intensity range [0, 255], and $\alpha = \beta = \gamma = 1$. Ideally, the maximum of SSIM($\mathcal{I}$) is 1, and the better quality of the detected seam corresponds to the higher value of SSIM($\mathcal{I}$).

As shown in Fig. 6, we show the seam location and quantitative contrast with other advanced seamline detection approaches to validate the superiority of our proposed approach in HSI stitching. One may find that all the methods can eliminate ghosting and blurring artifacts. However, the seams detected by the Seam-driven [50] and Farse [40] methods are very irregular and face the problem of passing through the salient white building. In the DP [64] and EDP [65] methods, the starting point and endpoint of the seams are not fixed, causing visible misalignments at the boundary of the overlap region. In Fig. 6(g) and (h), the detected seams in 5/2-normed [55] and our methods eliminate visible boundaries to some extent and located in the highly similar texture positions. Quantitatively, we estimate these derived seams by the above-mentioned SSIM index, which can take on an objective assessment. We can find that the SSIM index of our detected seam is 0.9371, which obtains the highest score in all the methods. Hence, our proposed method is capable of selecting the seam that meets the human visual perception well and derives a more quantitative evaluating index.

C. Evaluations on Smoothness Term

The ablation analysis of different energy costs is analyzed in this section. We analyze the effect on the obtained seam results when considering different spatial or spectral information at different costs. For the sake of fairness and precision, we keep the matching and alignment strategies consistent, only the difference costs defined in smoothness terms are changed, and the corresponding seam results are derived. The costs used for comparison include the color difference, the gradient difference, and the composite of color and gradient difference.

A pair of adjacent HSIs (i.e., 13th and 14th images) in our dataset are selected for this experiment. As illustrated in Fig. 7, the input images, warped images, difference energy costs, and derived seams in overlap regions are exhibited. Similarly, we calculated the SSIM index for each detected seam. If we only take on the color difference or the gradient difference, it is not hard to find that the obvious objects in energy maps are not extrusive enough, and the detected seam will cross the edge of the salient object. Although the composite of color and gradient difference derives a seam with better visual effects and avoids the visible objects, the cost of salient objects in the energy map is still not obvious enough. In comparison, our energy cost can highlight the visible objects better, and the corresponding seam is far away from these visible objects. Quantitatively, the SSIM index of our seam is higher, which proves that our proposed energy cost considering both the spatial and spectral information is more favorable to derive a higher quality seam for HSI stitching.

D. Spectral Performance Validation

As special data with higher dimensions than RGB images, HSIs, which possess abundant spectral characteristics, have been widely applied in many fields. For each pixel in the HSI stitching result, how to keep the uniform spectral information of the pixels of the same objects in the original image is a vital and inevitable problem. Thus, the analysis of spectral consistency is an essential quality indicator of the stitching performance in this work.

Here, several quantitative indices are used to verify the spectral performance. The Euclidean distance (ED) and root mean square error (RMSE) are two popularly adopted metrics to judge the deviation between two target spectra. Moreover, spectral angle distance (SAD) [67] represents the spectral similarity of two spectra by evaluating the angle divergence between two spectra. The formula of SAD is given by

$$\mathrm{SAD} = \arccos\left[\frac{\sum_{i=1}^{n} a_i b_i}{\sqrt{\sum_{i=1}^{n} a_i^2}\sqrt{\sum_{i=1}^{n} b_i^2}}\right]. \tag{15}$$

The spectral information divergence (SID) [68] is the other popular index to describe the spectral similarity of two spectra, which can be defined as follows:

$$\mathrm{SID} = \sum_{i=1}^{n} a_i \log\left(\frac{a_i}{b_i}\right) + \sum_{i=1}^{n} b_i \log\left(\frac{b_i}{a_i}\right) \tag{16}$$

where n is the total number of bands of HSIs, and a_i and b_i are the gray values of the test spectrum and reference spectrum, respectively. The smaller values of the SAD and SID indices represent the higher similarity of the two spectra.

To estimate the spectral consistency of each pixel in the overlap region in the composite image, the overlap region of a typical case with abundant terrain context (i.e., the eighth and ninth images) in our dataset is selected. Specifically, four representative types of land cover or obvious objects labeled as points A–D, respectively, corresponding to tree, road, building, and white car, are chosen from the above-mentioned area. As can be seen in Fig. 8, point A is located above the detected seam and points B–D are located below the detected seam. Ideally, the spectrum curve of each pixel is homologous to the pixel of the same position in the reference image to which it belongs [i.e., point A corresponds to the ninth warped reference image, points B–D correspond to the eighth warped reference image, the eighth and ninth warped reference images are expressed in Fig. 6(b)]. Fig. 9 shows

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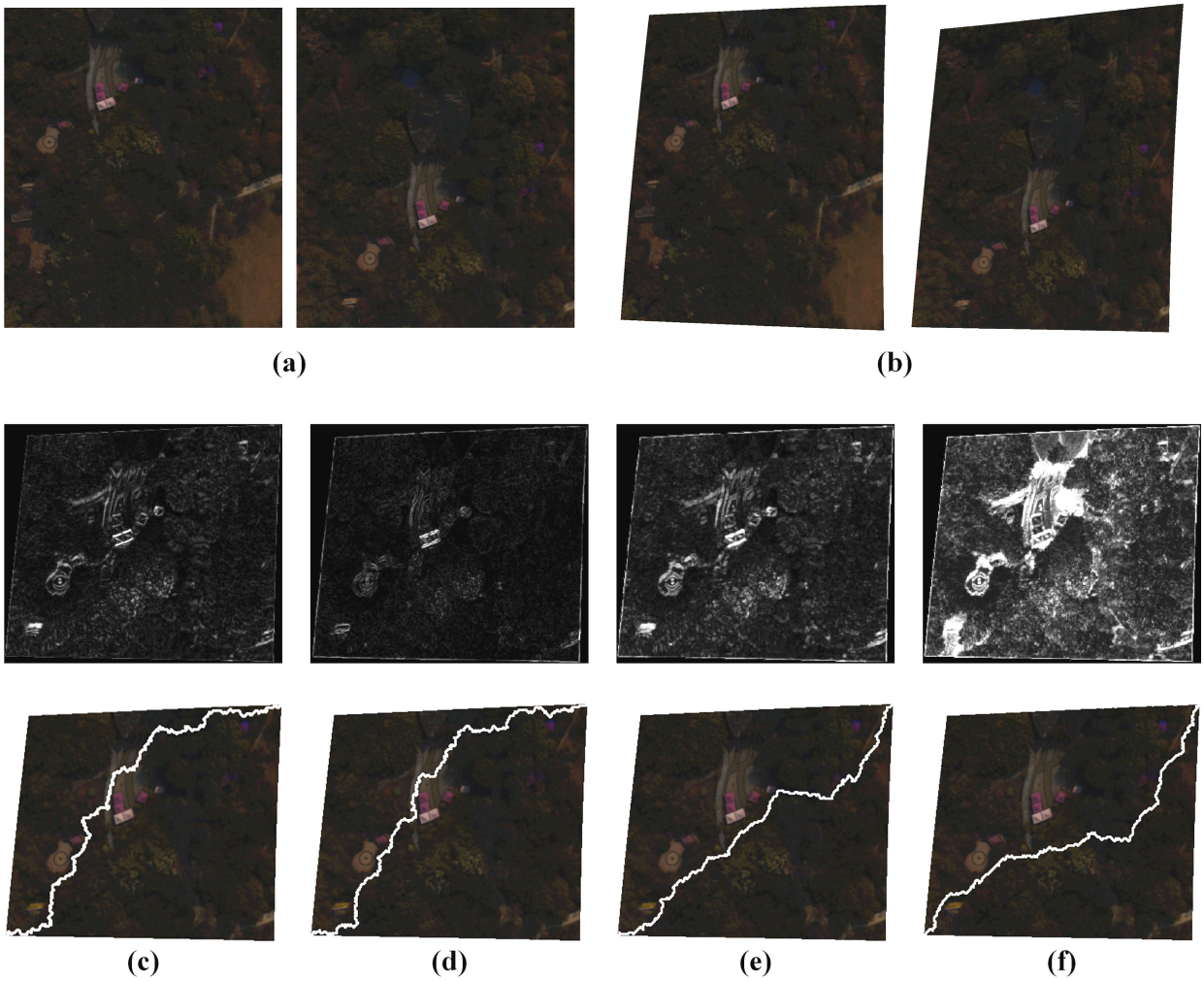


Fig. 7. Ablation analysis of the smoothness energy cost. From (c) and (f), (top) the energy maps in the overlap region, and (bottom) the corresponding seams. (a) Input images. (b) Warped images. (c) Color difference, SSIM index is 0.8792. (d) Gradient difference, SSIM index is 0.8757. (e) Combination of color and gradient difference, SSIM index is 0.9102. (f) Ours, SSIM index is **0.9258**.

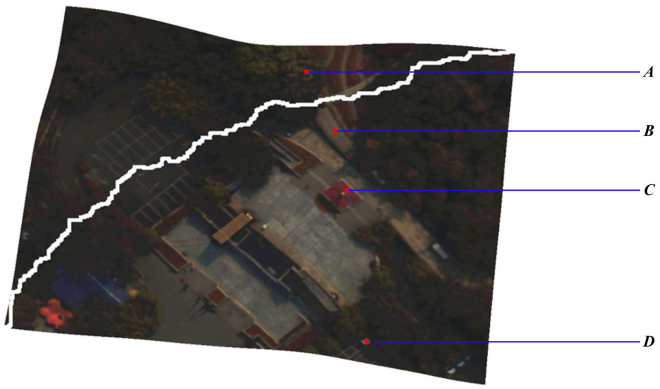


Fig. 8. Four typical points in the overlap region of aligned pair images.

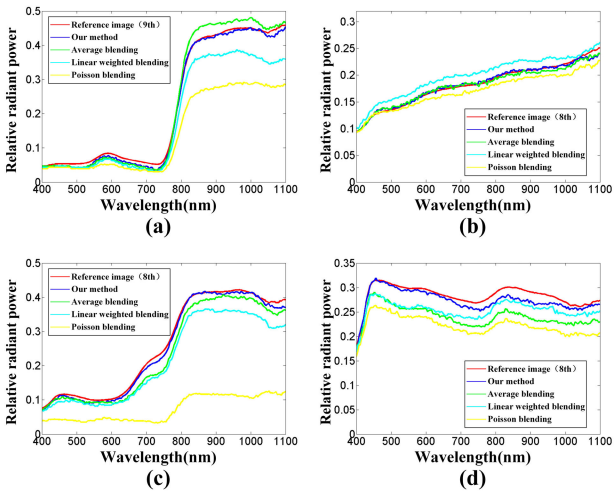


Fig. 9. Quantitative comparison of the spectrum stitching performance of our method and several blending methods. (a)–(d) Four selected typical points *A*–*D* in Fig. 7, respectively. (a) Tree. (b) Road. (c) Building. (d) White car.

the spectrum analysis results of our proposed strategy with three popular blending methods: AB [30], LWB [31], and PB [32]. One can find that the spectra of the pixels of our method are closer to the same position in the reference image. The detailed quantitative experimental results are listed in Tables I–IV. Lower ED, RMSE, SAM, and SID values verify that the stitching results of our proposed method have better spectral consistency, which illustrate that our method is able to combat the spectral distortion and ensures the integrity and consistency of the spectral information.

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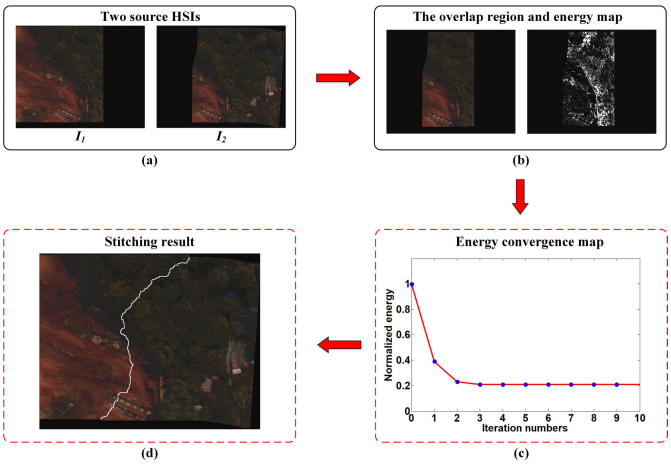


Fig. 10. Empirical convergence analysis of energy function solution. (a) Two source HSIs. (b) Overlap region and energy map. (c) Energy convergence map. (d) Stitching result.

TABLE I

QUANTITATIVE SPECTRAL CONSISTENCY ANALYSIS FOR POINT *A* (TREE), BOLD INDICATES THE BEST

	AB	LWB	PB	Ours
ED	0.1751	0.2239	0.3172	**0.0561**
RMSE	0.0133	0.0251	0.0428	**0.0056**
SAD	0.0331	0.0428	0.0491	**0.0267**
SID	0.0609	0.1008	0.1913	**0.0537**

TABLE II

QUANTITATIVE SPECTRAL CONSISTENCY ANALYSIS FOR POINT *B* (ROAD), BOLD INDICATES THE BEST

	AB	LWB	PB	Ours
ED	0.0393	0.0671	0.0701	**0.0287**
RMSE	0.0049	0.0057	0.0091	**0.0039**
SAD	0.0201	0.0239	0.0211	**0.0189**
SID	0.0453	0.0711	0.0699	**0.0227**

TABLE III

QUANTITATIVE SPECTRAL CONSISTENCY ANALYSIS FOR POINT *C* (BUILDING), BOLD INDICATES THE BEST

	AB	LWB	PB	Ours
ED	0.1325	0.1731	0.5382	**0.0573**
RMSE	0.0139	0.0261	0.0619	**0.0063**
SAD	0.0538	0.0461	0.1632	**0.0226**
SID	0.0747	0.0961	0.2981	**0.0433**

TABLE IV

QUANTITATIVE SPECTRAL CONSISTENCY ANALYSIS FOR POINT *D* (WHITE CAR), BOLD INDICATES THE BEST

	AB	LWB	PB	Ours
ED	0.1129	0.1695	0.2273	**0.0708**
RMSE	0.0238	0.0301	0.0367	**0.0178**
SAD	0.0389	0.0291	0.0509	**0.0263**
SID	0.0511	0.0591	0.0616	**0.0367**

E. Empirical Convergence Analysis

After the energy function is defined, the graph cuts model can be used to derive the optimization solution. The graph cuts algorithm is a very classical method, which has been validated to converge to the optimization within ten iterations [51]. Fig. 10 shows the empirical convergence analysis of our stitching process. Fig. 10(a) is the input two source HSIs, and Fig. 10(b) shows the overlap region and the obtained energy map from our energy function. After that, the graph cuts algorithm is used to derive the optimization solution, and the energy convergence map can be found in Fig. 10(c). It is not hard to find that after only the first iteration, the energy value is significantly reduced compared with the initial state, and after three iterations, the energy value tends to be stable and no longer changes. This phenomenon is consistent with the result in [51], which can demonstrate the effectiveness and convenience of our proposed method. From the solution by graph cuts, we can obtain the optimal seam and achieve seamless stitching result.

V. CONCLUSION

Visual artifacts and spectral distortion are challenging problems in HSI stitching. The traditional optimal seamline detection methods can effectively combat the artifacts in RGB images but are not suitable for HSI stitching. In this work, we introduce a novel optimal seamline detection strategy via graph cuts for synthesizing seamless UAV HSI. First, with the combination of feature matching and robust warping, the geometrical transformation of multiple adjacent images is estimated, and the adjacent HSIs have been aligned into the same coordinate system. Then, a novel energy function that considers both the spatial and spectral information is established to combat with the artifacts in the overlap region. Specifically, we aggregate the information of color difference, gradient difference, and spectral similarity into the smoothness term of our energy function. Finally, the graph cuts algorithm is solved so that we could effectively derive the perceptually optimal seam passing through highly texture similar positions and avoiding visible objects. This work solves the dilemma that the optimal seamline detection method cannot be directly used for HSI stitching and derives an optimal seamline in the whole HSI for the first time. A series of experiments on a real HSI dataset have been conducted, and the results indicate that our approach outperforms other state-of-the-art stitching methods in both the spatial and spectral performance.

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